

1. System State Model

The C2 link can be modeled as a continuous-time stochastic process with state $S(t)$ at time t :

- S_0 (**Nominal**): Link meets all RCP requirements (Transaction Time, Continuity, Availability, Integrity).
- S_1 (**Degrading**): Performance drops below static thresholds; impending loss (analogous to the start of the LC2L Decision Time).
- S_2 (**Lost**): LC2L state declared (Transponder 7400); Segment 1 & 2 contingency maneuvers are active.
- S_3 (**Recovering**): C2 link is re-established, and coordination with ATC is underway to exit the contingency mode.

1a. Canonical LC2L Event Timeline

The state model above is anchored to the standardised LC2L timeline, which defines four epochs. All KPIs in ME-02 are measured against these:

Epoch	Event	State entered
—	C2 link performance begins to degrade	S_1
T_0	LC2L Decision Time elapses; LC2L state declared, UA squawks 7400	S_2
T_1	Segment 1 ends, Segment 2 begins	S_2 (cont.)
T_2	C2 link is re-established	S_3
T_3	RP/ATC coordination complete; LC2L event over	S_0

Segment 1 ($T_0 \rightarrow T_1$) is a continuation of the current route, deliberately predictable so ATC and surrounding traffic can acknowledge the LC2L aircraft and manoeuvre around it.

Segment 2 ($T_1 \rightarrow T_2$) may be a continuation, a diversion, or a hold. The $T_2 \rightarrow T_3$ interval is RP/ATC coordination to return the UA to its nominal flight plan.

ICAO RPASP-16/WP-15 specifies the LC2L Decision Time as **30 s in terminal airspace** (within 30 NM of departure/arrival airport) and **240 s en route**. This is a standardised bound, not a free parameter, and any sweep of $\tau_{decision}$ must bracket it.

2. Mapping to the $\Phi\Lambda E\Pi$ Trapezoid

To quantify the temporal behavioral approach, we adapt the power engineering metrics to the C2 communication paradigm:

- **Φ (Robustness / Pre-disturbance):**
 - **Metric:** Time to first failure ($TTFF$) or the resilience of the link to stay in S_0 .
 - **Relation to RCP:** Governed by **Availability** and **Integrity**.
- **Λ (Degradation Phase / Disturbance Progress):**
 - **Metric:** The transition rate from $S_1 \rightarrow S_2$. This is bounded by the **LC2L Decision Time** ($\tau_{decision}$).
 - **Relation to RCP:** Governed by **Continuity** and **Transaction Time** degradation.
- **E (Degraded State / Loss):**
 - **Metric:** The duration the system remains in S_2 (Contingency Execution Time). We define the maximum survivable time ($\tau_{survive}$) before a catastrophic failure (e.g., fuel exhaustion or airspace violation).
- **Π (Recovery Phase):**
 - **Metric:** Time to restore the system from $S_3 \rightarrow S_0$, including ATC coordination time (τ_{ATC}).

3. Resilience Function $R(t)$

We define the system resilience $R(t)$ as the probability that the C2 link can safely recover from a disruption within the operational functional boundary:

$$R(t) = \int_0^{\tau_{survive}} P(S(t) = S_0 | S(t - dt) = S_3) dt$$

Where the overall system resilience index (RI) over a mission duration T can be evaluated by integrating the performance capacity $C(t)$ of the C2 link:

$$RI = \frac{1}{T} \int_0^T C(t) dt$$

Where $C(t) = 1$ in S_0 , and decays during S_1, S_2, S_3 .